



Software Requirements Specification

Aid: The Verifiable AI Research Assistant

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1. Introduction

This document provides a detailed specification of the requirements for the software project "Aid". Aid is envisioned as a next-generation, AI-powered research assistant designed to streamline the academic and professional research workflow. This SRS serves as the foundational agreement between stakeholders and the development team, outlining all functionalities, constraints, and quality attributes of the system.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a comprehensive and unambiguous description of the features and capabilities of the Aid platform. It is intended for a diverse audience, including project managers, software developers, quality assurance engineers, and other stakeholders. This document will guide the design, development, and testing phases of the project, ensuring that the final product aligns with the project's vision and user needs

1.2 Project Scope

Aid will be a web-based Software-as-a-Service (SaaS) application that empowers users to build, manage, and analyze their personal research libraries. The core value proposition of Aid is its commitment to **verifiability and transparency**. Aid will allow users to upload various document types (PDFs, web pages, etc.), interact with them through a conversational AI interface, and receive insights that are directly and explicitly linked back to the source material.

Key features within the scope include:

- A centralized, user-managed research library.
- Multi-document conversational analysis ("chat with your data").
- A "Source Highlighting" feature for instant verification of AI-generated claims.
- Semantic citation search and management.
- Collaborative workspaces for team-based research.
- Integration with external academic databases and reference managers.

The initial project will focus on delivering a robust Minimum Viable Product (MVP) encompassing the core research and analysis workflow, with a plan for future evolution to include more advanced features like multimedia analysis and specialized AI agents.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
Aid	The name of the software product described in this document.
AI	Artificial Intelligence.
API	Application Programming Interface.
CRUD	Create, Read, Update, Delete. Basic data manipulation operations.
LLM	Large Language Model. The core AI technology for processing and generating text.

MVP	Minimum Viable Product. A version of a new product which allows a team to collect the maximum amount of validated learning about customers with the least effort.
OCR	Optical Character Recognition. Technology to convert images of text into machine-encoded text.
PDF	Portable Document Format.
SaaS	Software as a Service.
SRS	Software Requirements Specification.
UI/UX	User Interface / User Experience.

2. Overall Description

2.1 Product Perspective

Aid is a new, standalone web application that enters a growing market of AI-powered research tools. While tools like ChatGPT offer general conversational AI, and others like Elicit or Research Rabbit focus on specific niches like data extraction or literature discovery, Aid aims to provide a holistic and trustworthy research environment. Its primary differentiator is the unwavering focus on source traceability, allowing users to trust the AI's output by providing direct, clickable links to the evidence within their own documents.

The system will be architected as a multi-tenant SaaS platform, relying on external, third-party LLM APIs (e.g., OpenAI, Anthropic) for its core intelligence, while building its unique value through its interface, workflow management, and data handling capabilities.

2.2 Product Functions

The major functions of the Aid platform will include:

- **Library Management:** Users can create a personal knowledge base by uploading documents (PDFs, DOCX), saving web pages, and organizing them into logical collections or projects.
- **AI-Powered Analysis:** Users can ask natural language questions about a single document or an entire collection. The AI will synthesize information, compare and contrast sources, identify themes, and answer specific queries.
- **Source Verification:** Every piece of information generated by the AI will be accompanied by a "Source Highlight" citation, allowing the user to instantly click and view the exact passage in the original document from which the information was derived.
- **Annotation and Note-Taking:** Users can highlight text, add comments, and create structured annotations within their documents, integrating their own insights with the AI's analysis.
- **Citation and Referencing:** The system will assist in finding and formatting citations in various academic styles (e.g., APA, MLA, Chicago).
- **Collaboration:** Teams can share workspaces, documents, and conversations, enabling collaborative research and analysis.
- **User Account Management:** Standard user registration, login, profile management, and subscription management functionalities.

2.3 User Characteristics

The target users for Aid are individuals involved in knowledge-intensive work who need to process and synthesize large amounts of textual information. Based on analysis, this includes:

- **Academic Researchers & PhD Students:** For literature reviews, manuscript writing, and staying current with research.
- **University Students (Undergraduate & Graduate):** For writing essays, dissertations, and completing research-based assignments.
- **Scientists & Physicians:** For reviewing clinical trial data, research papers, and medical literature.
- **Financial Analysts & Consultants:** For analyzing market reports, financial statements, and industry white papers.
- **Data Scientists & Engineers:** For reviewing technical documentation and research papers on new methodologies.

Users are expected to be tech-savvy but are not required to have any programming or AI expertise. The interface must be intuitive and accessible to a non-technical audience.

2.4 General Constraints

- The system must be developed as a web application accessible through modern web browsers.
- The system will depend on third-party APIs for LLM processing. System performance and cost will be tied to these external services.
- User data privacy and security are paramount. All data must be encrypted in transit and at rest.
- The initial development will adhere to the technology stack defined in the Project Constraints section (see 4.2).
- The system must comply with copyright laws regarding user-uploaded content. The responsibility for having the right to use uploaded materials lies with the user.

3. Functional Requirements

This section details the specific functions the Aid system must perform. Each requirement is assigned a unique identifier for traceability.

3.1 Data Requirements

FR-D-01: User Account Data

FR-D-01.1

The system shall securely store user profile information, including a unique user ID, name, email address, and a hashed password.

FR-D-01.2

The system shall store user preferences, such as the chosen UI theme (light/dark), default citation style, and preferred AI model (if applicable).

FR-D-01.3

The system shall maintain the user's subscription status, including the current plan (e.g., Free, Pro, Team), billing cycle, and feature entitlements.

FR-D-02: Research Library Data

FR-D-02.1

The system shall allow users to upload files in various formats, initially supporting PDF, DOCX, and TXT. The system shall store these files securely, associated with the user's account.

FR-D-02.2

The system shall store metadata for each uploaded document, including original filename, file size, upload date, and document type.

FR-D-02.3

The system shall perform Optical Character Recognition (OCR) on uploaded PDFs that contain images of text to make them searchable and analyzable.

FR-D-02.4

The system shall allow users to save web pages by providing a URL. The system will fetch, parse, and store the main textual content of the page as a document in the user's library.

FR-D-03: Workspace and Collection Data

FR-D-03.1

The system shall allow users to create logical containers (e.g., "Collections" or "Folders") to organize their documents.

FR-D-03.2

The system shall support the creation of "Workspaces" for collaboration, allowing multiple users to share access to a common set of collections and documents.

FR-D-03.3

The system shall store permissions for each workspace, defining user roles (e.g., Owner, Editor, Viewer).

FR-D-04: Interaction and Annotation Data

FR-D-04.1

The system shall store the history of conversations (prompts and AI responses) for each document and collection, linked to the user who initiated them.

FR-D-04.2

The system shall store user-created annotations. Each annotation must include the highlighted text, its location in the source document (e.g., page number, coordinates), the annotation content (comment/note), and the creator's user ID.

FR-D-04.3

The system shall store data for AI-generated citations, including the formatted citation string, the selected style, and a link to the source document.

3.2 Interface Requirements

FR-I-01: Dashboard Interface

FR-I-01.1

Upon login, the system shall present a main dashboard displaying the user's recent documents, collections, and a primary action button to upload new content.

FR-I-01.2

The dashboard shall feature a persistent navigation sidebar providing access to the user's library, workspaces, account settings, and help resources.

FR-I-02: Document Viewer Interface

FR-I-02.1

The system shall provide a two-pane view for document interaction: one pane displaying the document content (e.g., a PDF rendering) and the other pane for the conversational AI chat interface.

FR-I-02.2

The document viewer shall include standard controls for navigation (page up/down, go to page), zoom (in/out), and search within the document.

FR-I-02.3

Users shall be able to select text within the document viewer to trigger contextual actions, such as "Summarize," "Explain," or "Add Annotation."

FR-I-03: Chat Interface

FR-I-03.1

The chat interface shall include a text input field for user prompts and a display area for the conversation history.

FR-I-03.2

AI-generated responses containing information from the source document(s) MUST display ";Source Highlights." These highlights shall be interactive; clicking on one shall scroll the document viewer to the exact passage and highlight it.

FR-I-03.3

The chat interface shall support special commands or "agents" (e.g., using `@` mentions like `@document1` or `@search\_web`) to direct the AI's context and behavior.

FR-I-04: External System Interfaces

FR-I-04.1

The system shall interface with third-party LLM provider APIs (e.g., OpenAI API, Anthropic API) to send prompts and receive generated text.

FR-I-04.2

The system shall interface with a payment gateway API (e.g., Stripe) to handle user subscriptions and billing.

FR-I-04.3

The system shall provide an API for importing reference libraries from external managers like Zotero and Mendeley.

3.3 Navigational Requirements

FR-N-01.1

The application shall have a clear and consistent global navigation structure, accessible from all pages, allowing users to easily move between the Dashboard, Library, Workspaces, and Settings.

FR-N-01.2

The system shall implement breadcrumb navigation within nested structures (e.g., Workspace > Collection > Document) to help users understand their location and navigate back up the hierarchy.

FR-N-01.3

All interactive elements, such as buttons, links, and form fields, shall have clear visual states (hover, focus, active) to provide feedback to the user.

FR-N-01.4

The system shall provide a global search functionality that allows users to search for documents, collections, and annotations across their entire account.

3.4 Personalization Requirements

FR-P-01.1

Users shall be able to set a default citation style from a predefined list (e.g., APA 7th, MLA 9th, Chicago) in their account settings. This style will be used for all auto-generated citations.

FR-P-01.2

Users (on applicable subscription tiers) shall be able to select their preferred underlying LLM from a list of available models (e.g., GPT-4, Claude 3.5).

FR-P-01.3

Users shall be able to switch between a light and dark theme for the user interface.

FR-P-01.4

The system shall allow users to customize the AI's response length (e.g., Short, Medium, Detailed) for chat interactions.

3.5 Transactional Requirements

FR-T-01.1

The system shall provide a secure user registration process, requiring email verification to activate an account.

FR-T-01.2

The system shall implement a secure login mechanism with support for password reset via email.

FR-T-01.3

The system shall allow users to view available subscription plans (e.g., Free, Pro, Team) and their associated features and limits.

FR-T-01.4

The system shall securely process payments for users wishing to upgrade to a paid subscription plan, using an external payment gateway. The system should not store raw credit card information.

FR-T-01.5

Users shall be able to manage their subscription from their account settings, including viewing billing history, changing plans, and canceling their subscription.

FR-T-01.6

The system shall allow users to export their data, including documents, annotations, and chat histories, in common formats (e.g., Markdown, DOCX, PDF).

4. Non-Functional Requirements

This section defines the quality attributes, constraints, and external interfaces that the system must adhere to.

4.1 Expected Quality

NFR-Q-01: Performance

NFR-Q-01.1

The web application's core pages (Dashboard, Library) shall load in under 3 seconds on a standard broadband connection.

NFR-Q-01.2

AI chat responses should begin streaming to the user interface within 5 seconds of a prompt submission for typical queries.

NFR-Q-01.3

Document uploads shall be processed (including OCR for a 10-page scanned PDF) and made available for analysis within 60 seconds.

NFR-Q-02: Reliability & Availability

NFR-Q-02.1

The system shall have a target uptime of 99.5%, excluding scheduled maintenance windows.

NFR-Q-02.2

The system shall implement robust error handling for external API failures (LLM, payment gateway) and provide clear feedback to the user.

NFR-Q-02.3

User data shall be backed up daily to prevent data loss in case of a catastrophic system failure.

NFR-Q-03: Usability

NFR-Q-03.1

The user interface shall be intuitive and follow established web design conventions. A new user should be able to upload a document and start a conversation with it in under 5 minutes without requiring a tutorial.

NFR-Q-03.2

The system shall be responsive and usable across a range of screen sizes, from desktop monitors to tablet devices.

NFR-Q-03.3

The system shall provide contextual help and tooltips for complex features.

NFR-Q-04: Security

NFR-Q-04.1

All communication between the client and server shall be encrypted using HTTPS (TLS 1.2 or higher).

NFR-Q-04.2

User passwords must be securely hashed using a modern, salted hashing algorithm (e.g., Argon2, bcrypt).

NFR-Q-04.3

The system must be protected against common web vulnerabilities, including Cross-Site Scripting (XSS), SQL Injection, and Cross-Site Request Forgery (CSRF), by following OWASP guidelines.

NFR-Q-04.4

The multi-tenant architecture must ensure strict data isolation between tenants. A user from one tenant must never be able to access data from another tenant.

NFR-Q-05: Scalability

NFR-Q-05.1

The system architecture shall be designed to scale horizontally to accommodate a growing number of concurrent users.

NFR-Q-05.2

The system should be able to handle a user library of up to 1,000 documents without significant performance degradation in search or retrieval operations.

4.2 Project Constraints

NFR-C-01: Technology Stack

As this project is for a web engineering course, a specific, modern technology stack is prescribed to balance development speed, scalability, and learning objectives.

Component	Technology	Justification
Frontend Framework	React.js	User-specified. Excellent for building interactive and dynamic user interfaces. React provides a component-based architecture, efficient rendering with the virtual DOM, and a strong ecosystem that supports building scalable, production-grade applications.
Programming Language	TypeScript	Adds static typing to JavaScript, reducing bugs and improving code maintainability, which is crucial for a project of this complexity.
Backend Framework	Node.js with Express.js	Allows for a unified JavaScript/TypeScript codebase across frontend and backend. Its non-blocking I/O is well-suited for handling concurrent API requests and real-time interactions.
Database	PostgreSQL	A powerful open-source relational database with strong support for complex queries, JSON data types (for storing annotations and settings), and robust multi-tenancy patterns.
Styling	Tailwind CSS	A utility-first CSS framework that enables rapid UI development without leaving the HTML, fitting well with the component-based nature of React.
Deployment/Hosting	Netlify/Vercel (Frontend) & Railway/Render (Backend/DB)	Vercel and Netlify offers seamless deployment for modern frontend applications. Railway or Render provide easy-to-manage, scalable hosting for the backend and

		database, with developer-friendly pricing.
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NFR-C-02: Budget and Timeline

NFR-C-02.1

The project is constrained by the academic semester timeline. An MVP must be demonstrable by the end of the course.

NFR-C-02.2

The operational budget is limited. The choice of hosting and third-party services must prioritize free or low-cost tiers for development and initial deployment.

NFR-C-03: System Evolution

NFR-C-03.1

The architecture must be modular to allow for future expansion. Future features may include:

- Support for multimedia files (audio, video) with transcription.
- Advanced AI agents for specific tasks (e.g., '@CreateFlashcards', '@SystematicReview').
- Deeper integration with tools like Obsidian and Notion.
- A public API for third-party developers.

5. Conclusion

This Software Requirements Specification document establishes a comprehensive foundation for the development of the Aid platform. By detailing the functional and non-functional requirements, it provides a clear roadmap for creating a powerful, user-friendly, and, most importantly, trustworthy AI research assistant. The project's success will be measured by its adherence to these requirements, particularly its ability to deliver on the core promise of verifiable, source-traceable AI insights. This document will serve as a living guide throughout the project lifecycle, ensuring all stakeholders remain aligned on the goals and scope of this ambitious endeavor.

